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Notifications

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14. Symmetric matrices and their diagonalization

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Note that because the eigenvalues of a real symmetric matrix are always complete, a real symmetric matrix is always diagonalizable. More, it turns out, is true for symmetric matrices.

Fun Fact 1 If **A** is a symmetric, real $n \times n$ matrix, its eigenvalues are real.

Two vectors **v** and **w** are **orthogonal** if their dot product is zero. The dot product of the two vectors **v** and **w** can be written as $\mathbf{v}^T \mathbf{w}$ using ordinary matrix multiplication.

Fun Fact 2 If **A** is a symmetric, real $n \times n$ matrix, the eigenvectors of any two distinct eigenvalues are orthogonal.

In particular, this means that we can find a basis for the eigenspaces that are all mutually orthogonal. This is because any repeated eigenvalue has a complete eigenspace, and there is an algorithm, called the Gram–Schmidt algorithm, that allows us to find an orthogonal set of vectors.

Why do we care about orthogonal vectors?

Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be a collection of eigenvectors of **A** that are pairwise orthogonal, that is

$$\mathbf{v}_i^T \mathbf{v}_j = 0 \quad \text{if } i \neq j.$$

Normalize these vectors so that each of them has length one, that is

$$\mathbf{v}_i^T \mathbf{v}_i = 1.$$

Once the vectors are normalized in this manner, they are said to be **orthonormal** .

Let **S** be the matrix whose columns are these orthonormal vectors. Then we say that **S** is an **orthogonal matrix** , and has the special property that $\mathbf{S}^{-1} = \mathbf{S}^T$. This is convenient because computing inverses is often computationally intensive, but the transpose is simple.

In particular, the diagonalization of the matrix **A** can be expressed as $\mathbf{A} = \mathbf{S} \mathbf{D} \mathbf{S}^T$.

Remark 14.1 If you use the MATLAB command `[S,D] = eig(A)` to find the eigenvectors of a matrix **A**, the eigenvectors it finds will be unit length, and orthogonal if possible.

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